

FIRST PROGRESS REPORT – March 12, 2026

Project: CFPB Complaint Analysis

Aim:

Model the real-world customer complaint data using NLP for sentiment analysis and HMM for insights from state transitions.

Dataset: CFPB complaints data from Feb 2024 to Feb 2026

Tasks Accomplished:

1. Data cleaning, EDA and complaint text preprocessing.
2. Experimenting with various NLP techniques – Bag of words, n-grams, VADER, finBERT, RoBERT-a models.
3. Computing perplexity, determining optimum number of topics and validation of unigram vs bigram for topic modeling using LDA.
4. Finalized on these fronts – VADER features for sentiment analysis, 50 topics modeled using bigrams and LDA creating a final dataset for NLP workflow.

Current Output: ~1.4M complaints with 25 validated NLP features

Immediate Next Steps:

1. As I am done with Phase-1 (NLP signals), I will validate these features using a baseline and a target variable.
2. I am planning to use the label 'Company response to consumer' which is readily available in the dataset due to absence of any clear churn-based binary classification flag.
3. There are 5 classes in this label, so I need to check distributions and class imbalances here.
4. Finalizing target definition and model optimization through hyperparameter tuning (grid search).

Challenges and Adjustments:

1. No ground truth churn labels

Original Proposal : Predict customer churn from complaint text

Issue : CFPB dataset lacks customer identifiers and churn outcomes

Adjustment : Refining the project scope as multi-class complaint triage based on sentiment signals.

2. No customer-level sequences for traditional HMM

Original Proposal : Use HMM for defining different customer satisfaction stages (for churn prediction)

Issue : CFPB has complaint ids but no customer ids, so cannot link different complaints

Adjustment : Refining the project scope to create synthetically create links between different complaints and model complaint escalation patterns using HMM instead of customer satisfaction states.

Overall plan for project completion:

The most crucial step of the project now is to create a valid target definition and use a multi-class supervised learning approach to train, validate and select best model based on features derived till now. Further, prioritizing model optimization – tuning and temporal validation (OOS and OOT validation). Since there are no complaint ids in the dataset, I will create synthetic links between existing complaints to simulate state transitions and then train HMM to model and predict customer complaint escalation patterns between the previously defined target classes. Since the HMM phase involves simulation, synthetic link creation and training the HMM, it requires meticulous design and more attention. As it is the last phase of my proposed project, I will try to accomplish it by end of next month, but it is also vital to document and create the final report in a detailed manner. Expected end goal for me is to deliver an intelligent complaint triage system combining supervised prediction with simulation-based escalation modeling which could assist in intervention and optimization strategies for businesses to resolve and prioritize the complaints to avoid escalation and monetary loss and maintain reputation. However, if HMM implementation for full dataset proves too complex in a constrained time, I intend to complete Phases 1-2 fully (supervised classification with temporal validation) and try HMM for a smaller subset as a learning exercise and document HMM results for that sample.